Vishnu Pulipaka

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Professional Summary

- Researcher interested in large language models: how they behave across versions and providers, how well they generalize to languages with little training data, and how efficiently they can actually be served once they leave the paper.
- My interests sit across large language models, natural language processing, multilingual NLP, and edge AI, and I'm drawn to questions where the obvious explanation turns out to be wrong.
- Trained originally in electronics and computer architecture, which is part of why the systems side of machine learning, not just the modeling side, is where I keep ending up.
- Skeptical of results that are only ever tested in English, since most of the world's languages don't get the luxury of hundred-million-example benchmarks.

Technical Skills

Programming Languages: Python, C++, Embedded C, SQL
Machine Learning & Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, XGBoost
NLP & LLMs: Retrieval-augmented generation, LoRA fine-tuning, prompt engineering, Sentence Transformers, tokenization, LangChain
Distributed & High-Performance Computing: SLURM, FSDP, DeepSpeed, CUDA, multi-GPU (A100) training
Systems for ML Inference: TensorRT, ONNX Runtime, C++ inference engines
Embedded Systems & Computer Architecture: Embedded C, computer architecture, digital logic design, microcontroller programming

Publications

Under Review

1. *Evolving Fairness in Large Language Models: A Longitudinal Multi-Benchmark Study Across Model Families and Versions*. Under double-blind review.
   - Release-level fairness drift audit framework applied to 12 model versions from 5 providers across 6 model families and 7 bias benchmarks (about 200K examples).
   - Toxicity declines slightly on average, but the 95% confidence interval spans zero; stereotype scores hold steady or worsen in 67% of version transitions.
   - Gemini-2.5-Pro shows one statistically significant stereotype regression (Cohen's $d = 0.64$, $p < 0.001$).
   - Findings replicate after excluding StereoSet and CrowS-Pairs, the two most criticized benchmarks (8 of 8 drift pairs consistent).
2. *Can Translation Bridge the Data Gap for Low-Resource Languages? A Case Study of Hindi and Telugu*. Submitted to the BabyLM Workshop.
   - Translates the 100M-word English BabyLM corpus into Hindi and Telugu using IndicTrans2, then trains monolingual and bilingual GPT-Wee and GPT-BERT models.
   - GPT-BERT consistently outperforms GPT-Wee across both languages on perplexity and linguistic benchmarks.
   - Models trained on translated data generalize worse than models trained on native, non-translated CC-100 data when tested on out-of-domain text, despite stronger in-domain perplexity.

Preprints

1. *How Fast Can Reward Models Score? A Systems Study of C++ and PyTorch Inference Runtimes for RLHF*. arXiv:2607.19712, 2026. [arXiv] [Code]
   - Built a native C++ inference engine on ONNX Runtime for RLHF reward-model scoring, benchmarked against PyTorch eager mode, torch.compile, and FastAPI on CPU and GPU.
   - On CPU, the C++ engine beats every PyTorch baseline by 1.7–1.9× with non-overlapping confidence intervals; on GPU, torch.compile wins at both the median and the tail (p95).
   - The CPU speedup traces to ONNX Runtime's graph execution, not to C++ itself: calling the same ONNX session from Python performs statistically indistinguishably from the native engine.
   - Batching strategy matters more than language or runtime choice: naive padding costs 5–8× throughput on CPU and 3.5–4× on GPU relative to length-aware bucketing.

Manuscripts in Preparation

1. *Monolingual vs. Multilingual Models for Dravidian Languages: A Controlled Comparison Across Telugu, Tamil, Kannada, and Malayalam*. Target: EACL 2027.
   - Controlled comparison of monolingual and multilingual GPT-2 models across all four Dravidian languages, trained on a corpus of about 36.4GB from CC-100, Wikipedia, and Samanantar.
   - Evaluated against MuRIL, IndicBERT v2, mBERT, and XLM-R baselines.
2. *Tokenizer Efficiency Across Languages: Telugu, Hindi, and English*. Target: MRL 2026.
   - Evaluates subword tokenization across the three languages to measure token inflation, computational cost, and accuracy-efficiency trade-offs for morphologically rich, low-resource languages.
3. *MultiModalFedCD: A Privacy-Preserving Federated Learning Framework Combining Capsule Endoscopy and Gene Expression Data for Crohn's Disease Diagnosis*.

- Federated learning framework combining capsule endoscopy imaging with gene expression data for privacy-preserving Crohn's disease diagnosis.

Professional Experience

4SS Software Solutions Jun 2022 – Jul 2024
Software Engineer India

- Built Python and Java microservices and REST APIs on AWS, using Redis caching to reduce average response latency by 25%.
- Developed Kafka-based event-driven services and asynchronous workers, improving scalability and reducing interservice processing time by 22%.
- Shipped AI-augmented features, including LLM-powered utilities and data-processing pipelines, automating workflows and improving developer productivity by 15%.
- Optimized complex SQL queries and refactored legacy services, resolving 40% of bugs and reducing post-release defects by 15%.
- Set up CI/CD pipelines with GitHub Actions to build, test, and deploy applications to AWS, reducing deployment time by 20%.
- Collaborated with senior engineers and QA teams to troubleshoot distributed-system defects and improve release readiness within Agile delivery cycles.

Projects

TensorRT Inference Optimization Engine — Compiled ResNet-50 to an FP16 TensorRT engine, achieving $3.55\times$ throughput and a $4.3\times$ P99 latency reduction with zero accuracy loss. [Code]

RAG Evaluation Platform — Benchmarks 3 retrievers against 3 LLM providers on faithfulness, accuracy, latency, and cost, with automated regression gates. [Code]

Telco Customer Churn (MLOps Pipeline) — XGBoost churn classifier with a quality gate blocking any retrain that regresses PR-AUC by more than 0.01 against the production model. [Code]

ML Feature Store & Train-Serve Skew Detection — PySpark feature pipeline enforcing train-serve parity, with skew detected via SHA-256 feature-matrix hashing. [Code]

FinSight-AI — Asynchronous multi-agent financial research system combining FinBERT sentiment, retrieval, and LLM-as-judge scoring. [Code]

VocalLytics — Voice-to-SQL system converting natural-language queries into guarded SQL over 3.4M orders. [Code]

Education

Montclair State University Sep 2024 – Dec 2025
Master of Science in Data Science, GPA: 3.87/4.0
Montclair, New Jersey

Sreenidhi Institute of Science & Technology Nov 2020 – May 2024
Bachelor of Technology in Electronics and Communication Engineering
Hyderabad, India

Certifications

AWS Certified Solutions Architect – Associate February 2026